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DESCRIPTIVE WRITTEN TEST

CSA1407

① Smart Home Automation system using intelligent agents :

Introduction :

An intelligent agent is a system that observes its environment through sensors & takes suitable actions using actuators to achieve specific objectives. In a smart home automation system, sensors detect motion, temperature, light & time, while actuators control light, air conditions. The four major types of intelligent agents are simple reflex, model-based, goal-based & utility-based agents.

i) Simple Reflex Agent :

A simple reflex agent selects action based on the current input condition action rules.

working in a smart home :

- * If motion is detected, turn on the lights.
- * If the temperature exceeds 28°C, turn on air conditioner
- * If an unauthorized person is detected, active the alarm.

Advantages : Simple, fast, & easy to implement.

disadvantages : It does not remember past events or learn user perferences. It make unsuitable decisions when the environment changes.

ii) Model-Based Agent :

A model based agent maintains an internal state of the environment & users previous observation to make decisions.

Working in smart home :

* Remembers whether a room is occupied.
* Tracks the current temperature & previous temperature changes.
* Checks whether the room is in sleeping mode or away mode before controlling devices.

Advantages : Handles partially observable environment & makes better decision using stored information.

disadvantages : Requires more memory & a model of environment

③ Goal-Based Agent :

A goal-based agent selects action that help achieve a specific goal.

Working in a smart home :

-* Achieves the goal of maintaining a comfortable room temperature.
-* Ensures that doors are locked when the home is unoccupied.
-* Turns off unnecessary light to reduce energy consumption.

Advantages : Can plan action & choose different paths achieve goal.

disadvantages : Planning may require additional computation & time.

(iv) Utility-Based Agent :

A utility-based agent evaluates possible action using a utility function selects action greatest overall benefit.

Working in a smart home :

* Balances comfort with electricity consumption.
* Adjusts air conditioning based on temperature, occupancy, & energy cost.
* Prioritizes security while avoiding unnecessary alarms.

Advantages : Balances multiple objectives & makes flexible decision when trade-off exists.

② Robot Navigating an unknown maze using UCS & IDS

A robot navigating an unknown maze must discover the available paths, avoid obstacles & reach the exit without knowing the complete layout in advance. Uninformed search algorithms explore possible states without using additional heuristic to knowledge about close a state is to goal.

Two important uninformed search startegies are UCS & IDS :-

i) Uniform cost search (UCS) :

Uniform cost search expands the node with the lowest total path cost from the starting position. It uses a priority queue to select the next node.

* place the starting position in the priority queue with cost zero.
* Remove the node with the lowest path cost.
* check whether it is exit.
* If it not exit, explore its unvisited neighboring position.
* Repeat until the exit is found or no reachable position remain.

Ex : If one route to the exit costs 12 units & another costs 7 units, UCS finds the lower-cost provided all step costs non-negative complied

Advantages :

* finds the least-cost path when step costs are non-negative.
* work well when different paths have different costs.
* is complete under standard conditions such as finite branching factor & positive minimum cost.

disadvantage :

* May explore many unnecessary positions.
* Can be slow in large mazes.

Time & space complexity : For a general search the e-UCS can require exponential time & space. Its complexity depends on the branching factor positive step cost.

② **Iterative Deepening Search (IDS)** :

Iterative Deepening search combines the space efficiency of depth-first search with the completeness of breadth-first search for unit-cost paths.

Working :

- It start with depth limit of zero.
- Search for the exit within that limit.
- If the exit is not found, increases the limit to one.
- Repeat the search with progressively larger limits.
- Stop when the exit is found.

ex: the robot first checks position at depth 0, then 1, depth 2, & so until it reaches exits.

Advantages :

- Uses much less memory than UCS.
- Is complete when the branching factor is finite.

Disadvantages :

- Repeats search on previously visited shallow nodes.
- Can take a long time in mazes with large branching factor.
- Does not guarantee the least-cost path when actions have different costs.

ime & space complexity .

- Time : $O(b^d)$
- Space : $O(bd)$